FIFA Money Printer — Project Summary

Goal

Predict the per-game outcome (home win / draw / away win) of international football matches — aimed at the 2026 World Cup — and turn calibrated probabilities into betting signals (1X2 **and** secondary markets) with a staked portfolio.

Data

- **data/results.csv** — ~49k international matches 1872–2026 (martj42, Kaggle). The backbone.
- **salimt Transfermarkt dump** — player market values, national-team membership, profiles, and **injuries** (dated spells). Used for the optional squad-value feature.
- Auxiliary: former_names, goalscorers, shootouts (mostly unused).
- (`data/` is git-ignored; regenerate via download.py + Kaggle.)

Modules (src/, 9 files)

Module	Role
wc_pipeline.py	Causal features: Elo, form (off/def split), H2H, Glicko; build_features/build_dataset
wc_squads.py	competitive_only filter; confederation tagging + MV-coverage filter
wc_market_value.py	Leakage-safe as-of join of a country value series onto matches
wc_squad_dataset.py	Builds the country squad-value series offline; SquadValuer for live overrides
dixon_coles.py	Dixon-Coles goals model -> scoreline matrix -> 1X2 + secondary markets
walk_forward.py	Walk-forward backtest, model registry (LR/HGB), compare_models, IS/OOS/gap
predict.py	MatchPredictor: LR+DC blend, secondary markets (blend-reconciled), absence override
portfolio.py	Devig (Shin) + EV + push-aware fractional-Kelly across any market
run_with_mv.py	One-command end-to-end runner
chemistry.py	Causal club-chemistry feature (built, tested NULL; reference tool)

Model features (12, all causal pre-match snapshots)

- `elo_diff` — competition-weighted Elo (the dominant signal)
- `is_neutral`
- Form (last 10): `form_pts`, plus offense/defense split `form_gf` / `form_ga`, home & away
- Head-to-head (pair-specific, shrunk): `h2h_home_winrate`, `h2h_home_gd`, `h2h_logn`
- `glicko_exp` — uncertainty-shrunk win expectancy (Glicko-1 rating + RD)
- (+ `mv_log_diff`, `mv_missing` when market value is enabled — opt-in)

The model: LR + Dixon-Coles blend

- **Logistic regression** on the 12 features (linear, calibrated, hard to beat here).
- **Dixon-Coles** goals model: per-team attack/defense + home edge by time-weighted MLE from goals, with the low-score correction. Gives the full scoreline matrix.
- **Blend** (0.5 each) is the production 1X2 model — LR and a goals process are decorrelated, so averaging captures both, and DC repairs the draw structure.

Validation (walk-forward, post-2000, rolling 8y train -> 6mo test blocks)

Each game scored out-of-sample by the model trained only on its past. Tracks OOS log loss, an in-sample reference + IS->OOS gap (overfit alarm), Brier, accuracy, folds-beating-Elo.

Performance — the ceiling-breaking progression

model	OOS log loss	notes
Pure Elo baseline	0.894	60.1% accuracy; never predicts a draw
LR (original, 6 feat)	0.8484	15/17 folds beat Elo
+ off/def form + H2H	0.8449	the one feature gain
+ Dixon-Coles blend	0.8419	fixes draws; unlocks secondary markets
+ Glicko-1 `glicko_exp`	0.8405	LR alone 0.8466; 16/17 folds beat Elo
+ **DC expanding history + ~5y decay**	**0.8393**	DC was data-starved by the rolling window

Net gain **0.8484 -> 0.8393** (~1.1%). The wins came from richer TARGET (goals model) and more DATA (Glicko ratings; DC on all history), never from a fancier model class. The model is **well-calibrated** (draw class 0.221 predicted vs 0.220 actual).

Trials ledger — what worked and what didn't

Every idea below was tested with a proper causal backtest (not intuition). The pattern is the headline result: **gains came from a richer target and more data; everything that tried to add "a new signal beyond team strength" failed**, because Elo already integrates whatever shows up in results.

Trial	Outcome
Offense/defense form split + head-to-head	✓ 0.8484 -> 0.8449
Dixon-Coles goals model, blended with LR	✓ 0.8419; fixes draws, unlocks secondary markets
Glicko-1 uncertainty (`glicko_exp`)	✓ 0.8405; 16/17 folds beat Elo
Dixon-Coles on expanding history + ~5y decay	✓ 0.8393 (current SOTA)
World Cup goal-scale (secondary markets)	✓ real, ~2.6 SE effect; sharpens totals/ handicap

Blend-reconciled secondary markets; devig + `vs_fair`	✓ betting-layer correctness
Gradient boosting (HGB) / EBM	✗ overfit; LR wins (signal-limited, not capacity-limited)
DC rho widening / bivariate Poisson	✗ rho not at bound; conditional goal corr ~0
Glicko-2 volatility	✗ sigma ~constant; worse than Glicko-1 in the blend
Confederation / host / competition-importance	✗ no measurable effect
Team-level H2H aggregates (mean/std)	✗ re-encode strength; nil
Closeness feature / adaptive LR-DC blend (for draws)	✗ both null
Pairwise "bogey team" beyond Elo; H2H post-processing	✗ corr +0.02; post-process hurts
Momentum / win-streak	✗ residual corr ~0; captured by Elo + form
Market value as a strength feature	⚠️ ~77% redundant with Elo; opt-in, marginal
Injury availability (historical)	⚠️ negligible backtest lift; real value is LIVE
Squad chemistry (causal, from transfer_history)	✗ orthogonal to strength but residual corr -0.007
Squad-value concentration (Gini / CV / top-1)	✗ distribution shape; residual corr ~0

Why we're at the ceiling — the error budget

40% of matches carry 68% of all log loss, in two near-irreducible buckets:

match type	% matches	% of total log loss	mean log loss
Draws	22%	37%	1.43
Upsets (favourite lost)	18%	31%	1.49
Clean favourite wins	60%	31%	0.44

The model is excellent on "chalk" and the residual lives in genuinely surprising events that need information no historical feature contains (lineups, motivation, one-offs).

What was ruled out (honest negative results)

- **Gradient boosting (HGB) and EBM** — both overfit; LR wins (signal-limited, not capacity-limited).
- **DC rho / bivariate Poisson** — rho not at bound; conditional goal correlation ~0 (no positive dependence for bivariate Poisson to model). DC already fits the draw structure.
- **Glicko-2** — volatility signal useless (sigma ~constant); worse than Glicko-1 in the blend.
- **Confederation / host / competition-importance features** — no measurable effect.
- **Team-level H2H aggregates** (mean/std) — re-encode strength; nil.
- **Market value** — ~77% redundant with Elo (corr 0.77); residual-after-Elo only +0.105 correlation with outcome. Opt-in, Europe-centric, marginal.

- **Injury availability** — historical backtest lift negligible (top-30 mean robust); real value is live (auto-excludes current injuries / manual absence override).
- **Draw fixes**: closeness feature (softmax already handles decay) and adaptive LR/DC blend (optimal weight ~flat by |elo|) — both null.
- **Pairwise "bogey team" effect beyond Elo** — corr(prior pair residual, current) = +0.017; when model and H2H disagree the MODEL is right 62% of the time. H2H post-processing hurts.
- **Momentum / win-streak** — residual corr ~0 (-0.02); already captured by Elo + form.
- **Squad CHEMISTRY** (causal, dated from transfer_history) — orthogonal to strength (first non-redundant idea!) but corr(chem_diff, residual) = -0.007; small-nation/few-clubs artifact, not a winning signal.
- **Squad-value CONCENTRATION** (Gini / CV / top-1 share) — distribution shape, also null (residual corr ~0). Pattern confirmed: orthogonal != predictive; Elo integrates everything that shows up in results.

One thing that DID test real beyond the model: **World Cup goal-scale** -- WC finals score ~8% more than DC predicts (validated, ~2.6 SE / 444 matches), so `wc=True` applies a 1.08 scaling that sharpens totals / correct-score (1X2 unchanged).

Prediction + betting layer

- `MatchPredictor.predict(home, away, neutral, unavailable={team: [players]})` -> 1X2 blend. Synthetic-row method (appends the fixture, re-runs build_features) so live and training features can never drift. `unavailable=` drops named players (absence override); `wc=True` applies the World Cup goal-scale.
- **Markets** (all from the Dixon-Coles scoreline matrix): 1X2, over/under, total-goals buckets (0-1 / 2-3 / 4+), Asian handicap +/-1 (with push). Secondary markets are derived from a matrix RECONCILED to the blend's better 1X2 -- so the handicap inherits the blend's calibration (0.8393) plus DC's "win by 1 vs by 2+" margin shape.
- `portfolio.py` — **devig** (Shin / multiplicative) gives the bookmaker's fair probability; `vs_fair` = p_model - fair flags genuine disagreement with the sharp market vs a vig illusion. **Push-aware** EV/Kelly (edge = p*o - (1-push)); fractional Kelly + exposure caps; any market.

How to run (from src/)

```
python walk_forward.py            # backtest (LR, no MV)
python run_with_mv.py             # backtest WITH market value (--inju
python dixon_coles.py             # DC vs LR vs blend
python predict.py                 # demo predictions (LR / DC / blend
python portfolio.py               # demo portfolio (edit fixtures + od
```

Honest limitations & next steps

- **Information ceiling**: ~0.84 log loss / 61% accuracy is good for 3-way football; further accuracy needs new information that predicts surprises, not more features.
- **1X2 on major matches is an efficient market.** The realistic edge is in (a) the secondary markets the goals model unlocks, and (b) lower-profile games — which is exactly where squad coverage thins out.
- **Open / blocking**: a historical odds dataset (opening + closing lines) is required to measure Closing Line Value, backtest the betting strategy, devig properly, and build a

model-vs-market blend (the legitimate "second opinion" — external info the model lacks).